

UNIVERSITY OF CALIFORNIA, DAVIS

BERKELEY • DAVIS • IRVINE • LOS ANGELES • MERCED • RIVERSIDE • SAN DIEGO • SAN FRANCISCO



SANTA BARBARA • SANTA CRUZ

STABLE ISOTOPE FACILITY
UNIVERSITY OF CALIFORNIA
387 N QUAD AVE
RM 1210, PLANT AND ENVIRONMENTAL SCIENCES BUILDING
DAVIS, CALIFORNIA 95616-8780
530-752-8100

DEPARTMENT OF PLANT SCIENCES
COLLEGE OF AGRICULTURAL AND
ENVIRONMENTAL SCIENCES

$\delta^{18}\text{O}$ by TC/EA-IRMS

Methodology

Solid $\delta^{18}\text{O}$ samples are analyzed on an Elementar vario PYRO cube elemental analyzer interfaced to an Elementar precisiON IRMS (Elementar Analysensysteme GmbH, Langenselbold, Germany). Samples are thermally decomposed to CO in a glassy carbon reactor filled with glassy carbon, graphite felt, and lamp black at 1450°C. CO is isolated from interfering N_2 by an adsorption trap, which is heated to release CO for analysis. After separation, an aliquot of the analyte gas is carried to the IRMS for measurement.

Calibration and Reporting of Stable Isotope Ratios

Multiple instances of at least four laboratory reference materials are interspersed with samples and used for finalizing sample data and assessing analytical quality.

Provisional values for the samples and references are based on reference gas peaks introduced to the IRMS with each sample. Post run corrections are as follows:

- Isotope variability through the course of the analytical run is corrected by regression of drift references and applied equally to samples and references.
- Isotope variability attributable to varying amount of oxygen is corrected using a regression of size references and applied equally to samples and references.
- Isotope values are normalized to accepted values using bounding isotopic references.
- Elemental totals are calculated based on IRMS peak area using size references as a calibration curve.

All lab references have been calibrated against international reference materials, including: IAEA-V9, IAEA-600, USGS-42, USGS-43, and USGS-35. The final delta (δ) values are expressed relative to VSMOW.

Quality control reference materials: Nylon, Alanine, Caffeine (IAEA-600), Sodium Nitrate (USGS-35)

Quality assessment reference materials: Cellulose, Chitin

Measurement Uncertainty

All reference materials analyzed with a sample set are listed on the References worksheet of the data file.

Precision (σ), based on replicate measurements of the quality control and assessment material(s), is expected to be better than 0.3‰ for well-sized samples.

The analytical range is 50-500 μgO . Data outside the range of the references are flagged in the results; those data should be interpreted cautiously.

Limit of quantification (LOQ): 50 μgO

Revision Date

February 24, 2024

Approved By

Richard Doucett



References

Brand, W. A.; Coplen, T. B.; Aerts-Bijma, A. T.; Böhlke, J. K.; Gehre, M.; Geilmann, H.; Gröning, M.; Jansen, H. G.; Meijer, H. A. J.; Mroczkowski, S. J.; Qi, H.; Soergel, K.; Stuart-Williams, H.; Weise, S. M.; Werner, R. A. Comprehensive Inter-Laboratory Calibration of Reference Materials for $\delta^{18}\text{O}$ versus VSMOW Using Various on-line High-Temperature Conversion Techniques. *Rapid Commun. Mass Spectrom.* **2009**, 23 (7), 999–1019. <https://doi.org/10.1002/rcm.3958>.

Fourel, F.; Martineau, F.; Lécuyer, C.; Kupka, H.-J.; Lange, L.; Ojeimi, C.; Seed, M. $18\text{O}/16\text{O}$ Ratio Measurements of Inorganic and Organic Materials by Elemental Analysis-Pyrolysis-Isotope Ratio Mass Spectrometry Continuous-Flow Techniques. *Rapid Commun. Mass Spectrom.* **2011**, 25 (19), 2691–2696. <https://doi.org/10.1002/rcm.5056>.

Glossary

$^{\circ}\text{C}$	degrees Celsius
‰	per mil; 1 ‰ is equivalent to 0.001 or 1 mUr
^{13}C	stable isotope of carbon; mass number of 13
^{15}N	stable isotope of nitrogen; mass number of 15
δ	delta notation for isotopic composition; in per mil (" ‰ ") or mUr; 1 ‰ equals 1 mUr
σ	standard deviation
Air	atmospheric N_2 ; primary reference for measurements of nitrogen isotopes
Alfalfa Flour	alfalfa flour reference material
Amaranth Flour	amaranth flour reference material
Bovine Liver	bovine liver reference material
C	carbon
Chitin	chitin reference material
CO	carbon monoxide
CO ₂	carbon dioxide
EA	elemental analyzer
Enriched Alanine	alanine reference material enriched in ^{15}N
GC	gas chromatograph
Glutamic Acid	glutamic acid reference material
IAEA	International Atomic Energy Agency
IAEA-600	caffeine prepared by W. Brand and R. Werner, Max-Planck-Institute for Biogeochemistry, Jena, Germany
IRMS	isotope-ratio mass spectrometry
Keratin	keratin reference material
LOQ	limit of quantification; minimum signal for analyte to meet required signal-to-noise ratio
mUr	milliurey; 1 mUr is equivalent to 0.001 or 1 ‰
N_2	nitrogen
NIST	National Institute of Standards and Technology
Nylon	nylon reference material
QA	quality assurance; overall laboratory measures to ensure measurement quality
QC	quality control; activities and procedures used to evaluate quality requirements
RSIL	USGS Reston Stable Isotope Laboratory, Reston, Virginia
USGS	United States Geological Survey
USGS40	L-glutamic acid prepared by H. Qi et al., RSIL, USGS

USGS41	L-glutamic acid enriched in ^{13}C & ^{15}N prepared by H. Qi et al., RSIL, USGS
USGS42	human hair (Tibetan) prepared by H. Qi et al., RSIL, USGS
USGS43	human hair (Indian) prepared by H. Qi et al., RSIL, USGS
USGS61	caffeine prepared by H. Qi, RSIL, USGS, and A. Schimmelmann, Indiana University, Bloomington, Indiana
USGS64	glycine prepared by H. Qi, RSIL, USGS, and A. Schimmelmann, Indiana University, Bloomington, Indiana
USGS65	glycine prepared by H. Qi, RSIL, USGS, and A. Schimmelmann, Indiana University, Bloomington, Indiana
VPDB	Vienna PeeDee Belemnite; primary reference for measurements of carbon isotopes

General Resources

Berglund M, Wieser ME. 2011. Isotopic compositions of the elements 2009 (IUPAC Technical Report). *Pure Appl Chem.* 83(2):397–410. doi: 10.1351/PAC-REP-10-06-02.

Brand WA, Coplen TB, Vogl J, Rosner M, Prohaska T. 2014. Assessment of international reference materials for isotope-ratio analysis (IUPAC Technical Report). *Pure Appl Chem.* 86(3):425–467. doi: 10.1515/pac-2013-1023.

Commission on Isotopic Abundances and Atomic Weights. c2007–2008. CIAAW/IUPAC. <http://www.ciaaw.org/index.htm>.

Coplen TB, Hopple JA, Böhlke JK, Peiser HS, Rieder SE, Krouse HR, Rosman KJR, Ding T, Vocke RD Jr, Revesz KM, et al. 2002. Compilation of minimum and maximum isotope ratios of selected elements in naturally occurring terrestrial materials and reagents. *US Geological Survey Water-Resources Investigations Report 2001-4222*. doi: 10.3133/wri014222.

Coplen TB. 2011. Guidelines and recommended terms for expression of stable-isotope-ratio and gas-ratio measurement results. *Rapid Commun Mass Spectrom.* 25(17):2538–2560. doi: 10.1002/rcm.5129.

de Groot PA, editor. 2004. *Handbook of Stable Isotope Analytical Techniques*. Vol. 1. Amsterdam (NL): Elsevier.

de Groot PA, editor. 2008. *Handbook of Stable Isotope Analytical Techniques*. Vol. 2. Amsterdam (NL): Elsevier.

Dunn PJH, Carter JF, editors. 2018. *Good Practice Guide for Isotope Ratio Mass Spectrometry*. 2nd ed. Bristol (GB): FIRMS. Available from: <http://www.forensic-isotopes.org/gpg.html>.

Fry B. 2006. *Stable Isotope Ecology*. New York (NY): Springer. doi: 10.1007/0-387-33745-8.

Meier-Augenstein W. 2017. *Stable Isotope Forensics: Methods and Forensic Applications of Stable Isotope Analysis*. 2nd ed. Hoboken (NJ): Wiley. doi: 10.1002/9781119080190.

Mook W, de Vries JJ, editors. 2001. Environmental Isotopes in the Hydrological Cycle: Principles and Applications. *IHP-V Technical Documents in Hydrology No. 39*. Paris (FR): UNESCO/IAEA; 6 vol. Available from: http://www-naweb.iaea.org/napc/ih/IHS_publication.html.

National Institute of Standards and Technology. Gaithersburg (MD): US Department of Commerce. <https://www.nist.gov/>.

Reference Products for Environment and Trade. c2009. Vienna (AT): International Atomic Energy Agency (IAEA). <https://nucleus.iaea.org/rpst/index.htm>.

Sharp Z. 2017. *Principles of Stable Isotope Geochemistry*. 2nd ed. Available from: https://digitalrepository.unm.edu/unm_oer/1/.